

ASSIGNMENT 1

Q1] a) 143.75 (decimal)

BINARY

$$\begin{array}{r} 2 \overline{) 143} \\ 2 \overline{) 71} \quad 1 \\ 2 \overline{) 35} \quad 1 \\ 2 \overline{) 17} \quad 1 \\ 2 \overline{) 8} \quad 0 \\ 2 \overline{) 4} \quad 0 \\ 2 \overline{) 2} \quad 0 \\ 1 \end{array}$$

$$\begin{array}{r} 0.75 \times 2 = 1.50 \quad 1 \\ 0.50 \times 2 = 1.00 \quad 1 \end{array}$$

$$= 1000111.11$$

OCTAL

$$\begin{array}{r} 8 \overline{) 143} \\ 8 \overline{) 17} \quad 7 \\ 8 \overline{) 2} \quad 1 \\ 0 \quad 2 \end{array}$$

$$0.75 \times 8 = 6.00 \quad (6)$$

$$= 217.6$$

HEXADECIMAL

$$\begin{array}{r} 16 \overline{) 143} \\ 16 \overline{) 8} \quad 15 = F \\ \underline{ 0} \quad 8 \end{array}$$

$$= 8F$$

$$= 8F.C$$

$$0.75 \times 16 = 12.00$$
$$(12 = C)$$

b) 11011.101 (binary)

DECIMAL

$$1 \times 2^0 = 1 +$$

$$1 \times 2^1 = 1 +$$

$$0 \times 2^2 = 0 +$$

$$1 \times 2^3 = 8 +$$

$$1 \times 2^4 = 16$$

$$\underline{27}$$

$$1 \times 2^{-1} = 0.5 +$$

$$0 \times 2^{-2} = 0 +$$

$$1 \times 2^{-3} = 0.125 +$$

$$= 0.625$$

$$= \underline{\underline{27.625}}$$

OCTAL

$$\begin{array}{r} 11 \mid 011 \\ 011 \mid 011 \\ \hline 3 \quad 3 \end{array}$$

$$= 33$$

$$\begin{array}{r} .101 \\ \hline 5 \end{array}$$

$$= \underline{\underline{33.5}}$$

HEXADECIMAL

$$\underline{11011}$$

$$1 \mid 1011$$

$$11 = B$$

$$\begin{array}{r} .101 \\ \hline 5 \end{array}$$

$$= \underline{\underline{1B.5}}$$

c) 453 (octal)

DECIMAL

$$\begin{aligned} 3 \times 8^0 &= 3 + \\ 5 \times 8^1 &= 40 + \\ 5 \times 8^2 &= 256 \\ &= 299 \end{aligned}$$

BINARY

$$\begin{aligned} 4 &= 100 \\ 5 &= 101 \\ 3 &= 011 \\ &= 100101011 \end{aligned}$$

HEXA

$$\begin{aligned} &\underline{100101011} \\ &1 \quad 2 \quad B \\ &= 12B \end{aligned}$$

d) 1F.C (hexa decimal)

BINARY

$$\begin{aligned} 1 &= 0001 \\ F &= 1111 \\ C &= 1100 \\ &= 11111.11 \end{aligned}$$

DECIMAL

$$\begin{aligned} 1 \times 16^1 &= 16 + \\ F \times 16^0 &= 15 \\ &= 31 \\ C \times 16^{-1} &= 0.75 \\ &= 31.75 \end{aligned}$$

OCTAL

$$\begin{aligned} &\underline{11111} \\ &3 \quad 7 \\ &= 37.6 \end{aligned}$$

DECIMAL

143.75

27.625

299

31.75

BINARY

100011.11

11011.101

100101011

11111.11

OCTAL

217.6

33.5

453

37.6

HEXADECIMAL

8F.C

1B.A

12B

1F.C

Q2) a) $(325)_5 = ?_{10}$

$$\begin{aligned} 3 \times 5^2 &= 75 + \\ 2 \times 5^1 &= 10 + \\ 4 \times 5^0 &= 4 \\ &= 89_{(10)} \end{aligned}$$

b) $32.23_{(4)} = ?_{10}$

$$\begin{aligned} 32 & \\ \hline 3 \times 4^1 &= 12 + \\ 2 \times 4^0 &= 2 \\ &= 24 \end{aligned}$$

$$\begin{aligned} .23 & \\ \hline 2 \times 4^{-1} &= 0.5 + \\ 3 \times 4^{-2} &= 0.1875 \\ &= 0.6875 \\ &= 14.6875_{(10)} \end{aligned}$$

c) $231_{(10)} = ?_7$

$$\begin{array}{r} 7 \overline{) 231} \\ 7 \overline{) 33} \quad 0 \\ 7 \overline{) 4} \quad 5 \\ \quad 0 \quad 4 \\ = 450_7 \end{array}$$

d) $189_{10} = ?_6$

$$\begin{array}{r} 6 \overline{) 189} \\ 6 \overline{) 31} \quad 3 \\ 6 \overline{) 5} \quad 1 \\ \quad 0 \quad 5 \\ = 513_6 \end{array}$$

5) ~~1 0 1 0 1 1~~
1

$$= 176415_{(8)}$$

c)

	1	1	1	
	5	7	8	A
	A	B	D	E
	1	0	3	6
			8	

$$= 553_6$$

5) a) 10101₂

BINARY	1	0	1	0	1
OPERATION	copy	$1 \oplus 0$	$0 \oplus 1$	$1 \oplus 0$	$0 \oplus 1$
GRAY	1	1	1	1	1

Gray = 11111

b) 01110₂

BINARY	0	1	1	1	0
OPERATION	copy	$0 \oplus 1$	$1 \oplus 1$	$1 \oplus 1$	$1 \oplus 0$
GRAY	0	1	0	0	1

Gray = 01001

6) a) 01101 gray

GRAY	0	1	1	0	1
OPERATION	copy	$0 \oplus 1$	$1 \oplus 1$	$0 \oplus 0$	$0 \oplus 1$
BINARY	0	1	0	0	1

Binary = 01001

b) 10101 gray

GRAY	1	0	1	0	1
OPERATION	copy	$1 \oplus 0$	$0 \oplus 1$	$0 \oplus 0$	$0 \oplus 1$
BINARY	1	1	0	0	1

= 11001

$$3a) \left(\frac{312}{20} \right)_n = (13.1)_n$$

$$312r = 3r^2 + r + 2$$

$$20 = 2r$$

$$13.1 = 1 + 3 + 1/r$$

$$\frac{3r^2 + r + 2}{2r} = r + 3 + 1/r$$

$$3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$3r^2 - 2r^2 - r - 6r + 2 - 2 = 0$$

$$r^2 - 5r = 0$$

$$r(r - 5) = 0$$

$$r = 0 \quad r = 5$$

$$r = 5$$

$$b) 264r + 136r = 433r$$

$$264 = 2r^2 + 6r + 4$$

$$136r = r^2 + 3r + 6$$

$$433r = 4r^2 + 3r + 3$$

$$2r^2 + 6r + 4 + r^2 + 3r + 6 \pm 4r^2 + 3r + 3$$

$$3r^2 + 9r + 10 \neq 4r^2 + 3r + 3$$

$$4r^2 - 3r^2 + 3r - 9r + 3 - 10 = 0$$

$$r^2 - 6r - 7 = 0$$

$$r^2 + r - 7r - 7 = 0$$

$$r(r + 1) - 7(r + 1) = 0$$

$$(r - 7)(r + 1) = 0$$

$$\underline{r = 7} \quad r = -1$$

$$c) 221505_8 = 12345_r$$

$$221505_8 = ?_d$$

$$\begin{aligned} & 2 \times 8^5 + 2 \times 8^4 + 1 \times 8^3 + 5 \times 8^2 + 0 \times 8^1 + 5 \\ &= 65536 + 8192 + 512 + 320 + 0 + 5 \\ &= 74565 \end{aligned}$$

$$12345_r = 74565$$

$$r^4 + 2r^3 + 3r^2 + 4r + 5 = 74565$$

$$r^4 + 2r^3 + 3r^2 + 4r + 5 - 74565 = 0$$

$$r^4 + 2r^3 + 3r^2 + 4r - 74560$$

Assuming $r=16$

~~$r=16$~~ ~~$r=2$~~

$$16^4 + 2 \times 16^3 + 3 \times 16^2 + 4(16) =$$

$$= 65536 + 8192 + 768 + 64$$

$$= 74560$$

$$= 74560 + 5 = \text{RHS}$$

$$\therefore \underline{\underline{r=16}}$$